



Full Length Article

Infectious Disease

Omitting Ciprofloxacin Prophylaxis in Patients Undergoing Allogeneic Hematopoietic Stem Cell Transplantation and Its Impact on Clinical Outcomes and Microbiome Structure



Haneen Daoud-Asfour¹, Israel Henig², Itai Ghersin³, Stav Rakedzon⁴, Anat Stern⁵, Milena Pitashny³, Tsila Zuckerman^{2,6}, Haggai Bar-Yoseph^{3,6,*}

¹ Department of Nephrology, Rambam Health Care Campus, Haifa, Israel

² Department of Hematology, Rambam Health Care Campus, Haifa, Israel

³ Department of Gastroenterology, Rambam Health Care Campus, Haifa, Israel

⁴ Department of Internal Medicine H, Rambam Health Care Campus, Haifa, Israel

⁵ Infectious Disease Unit, Rambam Health Care Campus, Haifa, Israel

⁶ Bruce Rappaport School of Medicine, Technion Israel Institute of Technology, Haifa, Israel

Article history:

Received 31 October 2021

Accepted 17 December 2021

Key Words:

Ciprofloxacin
Allogeneic hematopoietic stem cell transplantation
Graft-versus-host disease
Microbiome
Prophylaxis

A B S T R A C T

Fluoroquinolone prophylaxis during allogeneic hematopoietic stem cell transplantation (allo-HSCT) reduces bloodstream infections. However, this practice affects the gut microbiome and potentially increases dysbiosis, which is closely related to transplantation outcomes, and lower gastrointestinal (GI) tract acute graft-versus-host disease (GVHD). This study assessed the impact of omitting ciprofloxacin prophylaxis on GI GVHD, clinical outcomes, and microbiome composition in patients undergoing allo-HSCT. In this single-center, retrospective study comprising recipients of allo-HSCT performed between 2018 and 2020, routine ciprofloxacin prophylaxis (the exposure variable) was stopped in December 2018. The primary outcome was acute lower GI GVHD within 100 days post-transplantation; secondary outcomes were 1-year overall survival, nonrelapse mortality, relapse, and overall acute GVHD. Outcomes were compared using univariate and multivariate analyses and Kaplan-Meier/competing-risk analyses. Sequential stool samples were collected prospectively from a subpopulation of recipients, and the microbiome composition was analyzed. Seventy-five of the 129 patients (58.1%) received prophylactic ciprofloxacin treatment. Baseline characteristics did not differ between the 2 study groups: patients with ciprofloxacin prophylaxis and those without ciprofloxacin prophylaxis. The rate of lower GI GVHD also did not differ between the 2 groups (24% versus 18.5%; $P = .597$). None of the secondary outcomes was significantly different between the 2 groups in univariate, multivariate, and time-to-event analyses. In addition, microbiome analysis in a subpopulation of 22 patients did not reveal any significant between-group difference in alpha or beta diversity. Omitting prophylactic ciprofloxacin during allo-HSCT did not affect microbiome composition, lower GI-GVHD rate, or other significant clinical outcomes. The use of prophylactic antibiotics in this setting should be evaluated further.

© 2021 The American Society for Transplantation and Cellular Therapy. Published by Elsevier Inc. All rights reserved.

INTRODUCTION

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is widely used as a curative treatment for malignant and nonmalignant hematologic conditions. Although effective, allo-HSCT is accompanied by complications, among which graft-versus-host-disease (GVHD) and infections are the major causes of nonrelapse mortality (NRM) [1]. Fluoroquinolone prophylaxis during the prolonged neutropenia after allo-HSCT

has been shown to reduce the rate of bloodstream infections (BSIs) caused by gram-negative bacteria, making this approach the standard of care [2]. However, the benefit of this prophylaxis on mortality is controversial [3,4], which, along with emerging data demonstrating that this approach may increase the risk for BSI with antibiotic-resistant bacteria, has prompted a reevaluation of antibiotic prophylaxis [5–7].

Gut microbiota composition changes due to antibiotic prophylaxis have long been suspected to be associated with the risk for acute GVHD [8]. More recently reported data demonstrate a profound decrease in intestinal microbiome diversity during allo-HSCT, which has been associated with significant increases in NRM, acute GVHD (total and gastrointestinal [GI]),

Financial disclosure: See Acknowledgments on page 160.e7.

*Correspondence and reprint requests: Haggai Bar-Yoseph, MD, Department of Gastroenterology, Rambam Health Care Campus, 8th Haalia Hashnia St, Haifa 3109601, Israel.

E-mail address: haggaiy@gmail.com (H. Bar-Yoseph).

acute GVHD-related mortality, and overall survival (OS) [9–12]. One factor that may contribute to gut microbiota perturbation in this setting is fluoroquinolone prophylaxis, leading some centers to omit or modify transplantation antibiotic prophylaxis regimens in an effort to decrease microbiota insult and improve outcomes, specifically acute GI-GVHD [13,14]. In January 2019, the prophylactic ciprofloxacin given routinely to allo-HSCT recipients at our center was stopped. Here we describe the effect of omitting ciprofloxacin prophylaxis after allo-HSCT on acute lower GI GVHD, clinical outcomes, and microbiome composition.

METHODS

Patients and Study Design

This retrospective study included all adult patients (age ≥ 18 years) who underwent allo-HSCT at a tertiary hospital in Israel between February 2018 and June 2020. Patients were followed-up for 1 year after allo-HSCT or until death. Demographic and clinical data were retrieved from the electronic medical record and were supplemented by an electronic medical record database extraction tool (MDClone, Beer Sheva, Israel). In a nested cohort of patients, stool samples for microbiome analysis were collected prospectively (see below). The study was approved by the hospital's Institutional Review Board (0411-09-RMB and 0624-17-RMB) and was conducted in adherence to the principles of the Declaration of Helsinki.

Exposure Variable and Outcomes

Exposure variable

Routine prophylactic ciprofloxacin administration to allo-HSCT recipients at our center was stopped on January 1, 2019. Ciprofloxacin administration was considered as prophylactic when administered between February 1 and December 31, 2018, or in the period after this date when administered as treatment for infection during the period of neutropenia as a single antibiotic agent. Although serving another purpose, these patients were included in the ciprofloxacin prophylaxis cohort owing to similar and hypothesized effects on gut microbiome composition and diversity.

Outcomes

The primary outcome was the incidence of lower GI GVHD by day 100 post-transplantation. Secondary outcomes were OS, NRM, and relapse in the first year post-transplantation, as well as the incidences of acute GVHD and lower GI GVHD grade 2–4 by day 100 post-transplantation. Acute GVHD grading was assigned according to the MAGIC criteria [15]. BSIs with gram-negative bacteria and *Enterococcus* up to 14 days and total BSI up to 40 days served as secondary outcomes as well.

Microbiome Analysis

Stool collection and DNA extraction

Sequential stool samples were collected from a nested cohort of patients. The first sample was collected within 1 week before initiation of conditioning (pretransplantation sample). The second sample was collected periengraftment on day 16 (± 7 days) after transplantation (post-transplantation sample). In patients who developed lower GI GVHD, a third sample was collected after onset of GVHD at a median of 64 days (range, 29 to 187 days) post-transplantation. Fecal samples were frozen at -80°C within 6 hours of collection. DNA extraction was performed using the DNEASY PowerSoil Pro Kit (#47014; Qiagen, Germantown, MD) according to the manufacturer's recommendations. Quantification of DNA was done with a NanoDrop spectrophotometer (Thermo Fisher Scientific, Waltham, MA).

Shotgun metagenomics and downstream analyses

Shotgun metagenomics was performed using the CosmosID platform (Germantown, MD). A more detailed description of the sequencing and annotation is provided in Supplementary Data, Methods. Microbial communities were compared pretransplantation, post-transplantation, and after the development of lower GI GVHD. In the analysis of the post-transplantation microbiome, samples were grouped by the exposure variable (ciprofloxacin prophylaxis). The primary analysis was done using the R Phyloseq package [16]. After rarefaction to an even depth, alpha diversity indices (Chao1, Shannon, and Simpson) were compared between time points using the Kruskal-Wallis test along with Dunn's test for post hoc analyses, followed by the paired Mann-Whitney *U* test for the before-after transplantation comparison and the Mann-Whitney *U* test for comparisons between groups at a time point. Relative abundance was calculated and used for beta diversity. Beta diversity distance matrix (Bray-Curtis) was compared using the R vegan package function ADONIS [17], a multivariate analysis of variance based on dissimilarity with 999 permutations. A permutation-based implementation of Tukey's test was used to test pairwise differences between groups' beta dispersion. Differences in specific microbial taxa were assessed by LefSe

analysis on the Huttenhower lab Galaxy server (<http://huttenhower.sph.harvard.edu/galaxy/>), and the MaAsLin2 R package [18] was used to identify associations between clinical metadata and bacterial abundance matrix after centered log transformation of the data. The ALDEx2 R package [19] was used to look for a differential abundance of genes in the total matrix and for the analysis of antibiotic-resistant genes (ARGs) stratified by the exposure variable (ciprofloxacin prophylaxis). Results were visualized using the ggplot2 package [20]. Significance was set at a 2-sided *P* value $< .05$ after false discovery rate (FDR) correction for multiple comparisons using the Benjamini-Hochberg method. All analyses were performed using R statistical software (R Foundation for Statistical Computing, Vienna, Austria).

Statistical Methods

Univariate comparisons between patients who received ciprofloxacin prophylaxis and those who did not were performed using the chi-square test and Fisher's test for the categorical variables and the Mann-Whitney test for continuous variables, assuming non-normality. Multivariate logistic regression was used to look for confounders for the lower GI GVHD and total GVHD at 100 days after allo-HSCT. Variables entered into the regression included ciprofloxacin prophylaxis, conditioning protocol, donor type, and female donor to male recipient. The cumulative incidence of lower GI GVHD in the presence of death as a competing risk was performed, and testing of equality across groups by Gray's test was performed using the cmprsk R package [21]. Relapse and NRM were competing risks for each other. Kaplan-Meier curves were used to compare OS between groups, and the log-rank test was used to compare median survival. Multivariate Cox proportional hazard regression was used for prediction of 1-year OS and included age, ciprofloxacin prophylaxis, conditioning protocol, donor type, and indication for allo-HSCT. The R survival package [22] was used for fitting Cox models and generating Kaplan-Meier estimates. There were no missing data for these analyses.

RESULTS

Patient Population

A total of 129 patients who underwent allo-HSCT between February 2018 and June 2020 were included in our analysis. The median patient age was 54.5 years (range, 19 to 73 years). Acute myelogenous leukemia was the most common indication for allo-SCT (69%). All 129 patients received peripheral blood stem cell allografts, and in 127 patients the GVHD prophylaxis included cyclosporine from day -3 with methotrexate on days +1, +3, and +6. Sixty-seven patients (51.9%) received antithymocyte globulin (5 mg/kg/day for 3 days). There were no significant differences in baseline demographic and clinical characteristics between the study groups (Table 1).

Seventy-five patients (58.1%) received prophylactic ciprofloxacin, and the other 54 (41.9%) did not. The median duration of ciprofloxacin prophylaxis administration was 11 days (interquartile range, 7 to 17 days); however, the majority of patients received to broad-spectrum antibiotic therapy before or during transplantation. Almost all the patients (98.4%; 127 of 129) received at least 1 course of piperacillin/tazobactam or meropenem in the weeks following allo-HSCT (Table 1). In 109 patients (85.8%), this broad-spectrum antibiotic treatment was initiated by day 14 post-transplantation.

Omitting Ciprofloxacin Prophylaxis Was Not Associated with GVHD Incidence or Transplantation Outcomes

Twenty-eight of the 129 patients (21.7%) were diagnosed with lower GI GVHD within 100 days post-transplantation. There was no significant difference between patients who received prophylactic ciprofloxacin treatment and those who did not (24% versus 18.5%; *P* = .597). In the 2 groups, the incidence of grade II–IV acute GVHD of the lower GI was 12% and 12.9%, respectively (*P* = 1), and that of overall acute GVHD was 50.6% and 40.7%, respectively (*P* = .349). At 1 year post-transplantation, OS was 65.3% and 68.5%, respectively (*P* = .850). Relapse occurred in 5 patients (6.6%) in the prophylaxis group and in 8 patients (14.8%) in the no prophylaxis group (*P* = .224). The NRM rate was also similar in the 2 groups, at 33.3% and 27.7%, respectively (*P* = .631).

Table 1
Baseline Characteristics

Characteristic	No Ciprofloxacin Prophylaxis (N = 54)	Ciprofloxacin Prophylaxis (N = 75)	P Value
Females/males, n (% females)	24/30 (44.4)	33/42 (44)	1
Age, yr, mean (SD)	49 (15.6)	51.4 (14.5)	.58
Body mass index, mean (SD)	26.2 (5.5)	26.54 (5.1)	.24
HCT Comorbidity Index, mean (SD)	2.10 (1.96)	1.85 (1.77)	.65
Allo-HSCT indication, n (%)			
AML/MDS/CMML	41 (75.9)	48 (64)	
ALL	9 (16.6)	8 (10.6)	
Lymphomas	2 (3.7)	9 (12)	
Other*	2 (3.7)	10 (13.3)	
Intensity of conditioning, n (%)			.122
Myeloablative [†]	33 (61.1)	32 (42.6)	
Reduced intensity [‡]	9 (16.6)	24 (32)	
Reduced toxicity [§]	9 (16.6)	11 (14.6)	
Sequential	3 (5.5)	5 (6.6)	
Other	0 (0)	3 (4)	
Donor type, n (%)			.459
MUD	20 (37)	32 (42.6)	
MRD	24 (44.4)	33 (44)	
Haploidentical	0 (0)	2 (2.6)	
MMUD	10 (18.5)	8 (10.6)	
Female donor to male recipient, n (%)	8 (14.8)	10 (13.3)	.803
CMV serostatus, n (%)			.93
Donor negative/recipient negative	2 (3.7)	3 (4)	
Donor positive/recipient negative	4 (7.4)	4 (5.3)	
Donor negative/recipient positive	8 (14.8)	13 (17.3)	
Donor positive/recipient positive	39 (72.2)	47 (62.6)	
Antithymocyte globulin use, n (%)	29 (53.7)	37 (50.6)	.871
Piperacillin/tazobactam or meropenem administration during allo-HSCT, n (%)	52 (96.3)	75 (100)	1

Conditioning toxicity is considered the highest in myeloablative protocols resulting in increased risk for early morbidity and mortality. As intensity decreases, tissue damage is expected to decrease, and thus the reduced-toxicity and especially reduced-intensity protocols are associated with attenuated intestinal mucosal damage and transplantation-related mortality.

AML indicates acute myelogenous leukemia; MDS, myelodysplastic syndrome; CMML, chronic myelomonocytic leukemia; ALL, acute lymphoblastic leukemia; MUD, matched unrelated donor; MRD, matched related donor; MMUD, mismatched unrelated donor; CMV, cytomegalovirus.

* Other indications included aplastic anemia, chronic lymphoblastic leukemia, multiple myeloma, chronic neutropenia, myelofibrosis, and chronic myelogenous leukemia.

[†] Myeloablative protocols included busulfan + cytoxan and fludarabine + busulfan for 4 days.

[‡] Reduced-intensity protocols included fludarabine + busulfan for 2 days, fludarabine + melphalan, and BCNU + thiotepea + fludarabine.

[§] Reduced-toxicity protocols included treosulfan + fludarabine.

^{||} Received post-transplantation cyclophosphamide on days +3 and +4 with cyclosporine and mycophenolate mofetil from day +5.

(Table 2). Finally, there was no significant between-group difference in the rate of BSI in this limited cohort of patients (Table 2). Multivariate analysis including ciprofloxacin prophylaxis and other factors known to be correlated with the risk for acute GVHD identified female donor to male recipient as associated with an increased risk for lower GI GVHD (odds ratio, 3.29; 95% confidence interval, 1.03 to 10.3; $P = .04$), but ciprofloxacin prophylaxis was not (Table 3). However, this association was not evident for the overall risk of acute GVHD (Supplementary Table S1).

Accounting for death as a competing risk, the cumulative incidence of lower GI GVHD was not significantly different between the 2 groups ($P = .33$, Gray's test) (Figure 1). Kaplan-Meier analyses of OS also revealed no significant between-group difference ($P = .82$, log-rank test) (Figure 2). Furthermore, ciprofloxacin prophylaxis did not have a significant impact on the cumulative incidence of relapse and NRM rate at 1 year post-transplantation when analyzed as competing risks for each other ($P = .13$ for relapse and $P = .24$ for NRM, Gray's test) (Figure 3). Cox proportional hazard regression for 1-year

Table 2
Primary and Secondary Outcomes

Outcome	No Ciprofloxacin Prophylaxis (N = 54)	Ciprofloxacin Prophylaxis (N = 75)	P Value
Lower GI GVHD within 100 d post-transplantation, n (%)	10 (18.5)	18 (24)	.597
Grade 2–4 lower GI GVHD within 100 d post-transplantation, n (%)	7 (12.9)	9 (12)	1
GVHD within 100 days post-transplantation, n (%)	22 (40.7)	38 (50.6)	.349
Upper and lower GI, n (%)	17 (31.4)	29 (38.6)	
Skin, n (%)	9 (16.6)	16 (21.3)	
Liver, n (%)	4 (7.4)	4 (5.3)	
One-year OS, n (%)	37 (68.5)	49 (65.3)	.850
One-year NRM, n (%)	15 (27.7)	25 (33.3)	.631
One-year relapse rate, n (%)	8 (14.8)	5 (6.6)	.224
Total BSIs up to 40 d post-transplantation, n (%)	22 (40.7)	26 (34.7)	.603
Gram-negative bacteria and <i>Enterococcus</i> BSIs up to 14 d post-transplantation, n (%)	15 (27.8)	22 (29.3)	1

Table 3
Multivariate Analysis of Lower GI GVHD at 100 Days after Allo-HSCT

Characteristic	Odds Ratio	95% Confidence Interval	P Value
Ciprofloxacin	1.36	0.54–3.57	.5
Intensity of conditioning			
Myeloablative*	-	-	
Reduced intensity [†]	1.67	0.57–4.78	.3
Reduced toxicity [‡]	0.62	0.13–2.30	.5
Sequential [§]	0.51	0.03–3.40	.6
Other	2.23	0.10–26.7	.5
Donor type			
MUD	-	-	
MRD	1.03	0.38–2.77	>.9
Haploidentical	0		>.9
MMUD	1.79	0.44–6.84	.4
Female donor to male recipient	3.29	1.03–10.3	.04

Conditioning toxicity is considered the highest in myeloablative protocols, resulting in increased risk for early morbidity and mortality. As intensity decreases, tissue damage is expected to decrease, and thus the reduced-toxicity and especially the reduced-intensity protocols are associated with attenuated intestinal mucosal damage and transplantation-related mortality.

* Myeloablative protocols included busulfan + cytoxan and fludarabine + busulfan for 4 days.

[†] Reduced-intensity protocols included fludarabine + busulfan for 2 days, fludarabine, melphalan, BCNU, thiopeta, + fludarabine.

[‡] Reduced-toxicity protocols included treosulfan + fludarabine.

[§] Sequential protocols included FLAMSA (fludarabine, amsacrine, and cytarabine with 400 total body irradiation and cyclophosphamide).

mortality that included risk factors for mortality did not show an increased hazard related to ciprofloxacin prophylaxis (Supplementary Table S2).

Gut Microbiome Diversity Decreased after Allo-HSCT

A total of 50 stool samples were collected from 22 patients. Fifteen of these patients (68.2%) received ciprofloxacin prophylaxis, 8 (36.4%) developed lower GI GVHD and 9 (36.4%) died in the first year post-transplantation (Supplementary Table S3). After filtering 2 samples for low read depth (<0.9 million reads per sample), 48 samples were included in the analysis (20 pretransplantation, 20 post-transplantation, and 8 after the development of lower GI GVHD). A total of 155,542,365 reads (median, 3,110,847; range, 53,466 to 12,370,707) were sequenced. A total of 1276 hits were classified, 835 of them after filtering for rare taxa. For alpha diversity, the samples were rarefied for an even depth of 256,313 reads. The Chao1, Simpson, and Shannon indices were significantly different across collection time points ($P < .001$, $P = .02$, and $P = .02$, respectively, Kruskal-Wallis test). In a post hoc analysis, these indices were lower in the post-transplantation samples compared with the pretransplantation samples ($P < .001$, $P = .01$, and $P = .008$, respectively, Dunn's test; $P < .001$, $P = .015$, and $P = .0037$, respectively, paired Wilcoxon signed-rank test) (Supplementary Figure S1). For beta-diversity distance, Bray-Curtis distances are shown in Supplementary Figure S2A. Although beta dispersion was significantly

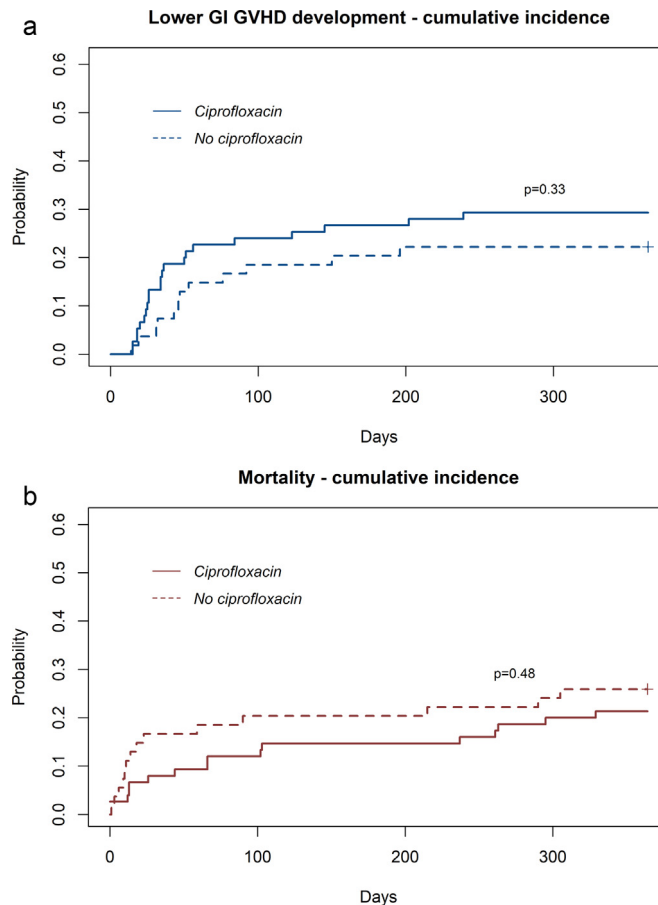


Figure 1. Cumulative incidence of lower GI GVHD and mortality using competing-risk analysis. (A) Cumulative incidence of lower GI GVHD development grouped by ciprofloxacin prophylaxis (solid line, ciprofloxacin; dashed line, no ciprofloxacin). (B) Cumulative incidence of mortality grouped by ciprofloxacin prophylaxis (solid line, ciprofloxacin; dashed line, no ciprofloxacin) within the first year after allo-HSCT. Results from Gray's test for each outcome are depicted.

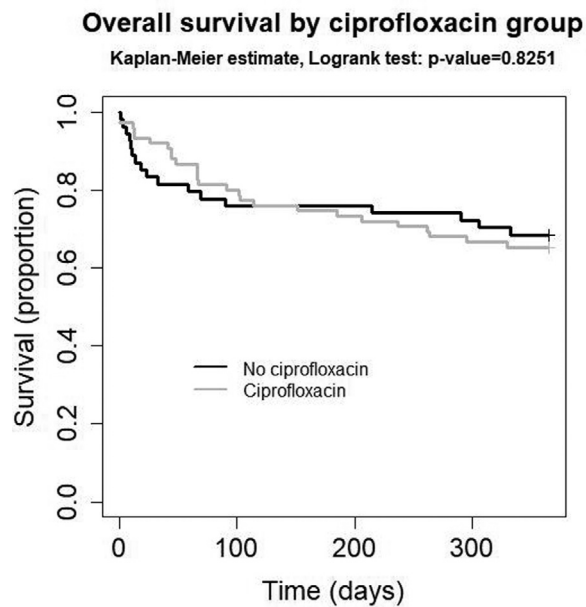


Figure 2. Kaplan-Meier survival curves for total mortality of patients who received ciprofloxacin prophylaxis (gray) and those who did not (black) within the first year after allo-HSCT. No significant difference was found between the study groups ($P = .825$, log-rank test).

different pretransplantation versus post-transplantation, it was not different in the samples analyzed after the development of GVHD (Supplementary Figure S2C).

Gut Microbiome Changes during Transplantation Were Not Associated with Ciprofloxacin Prophylaxis

Analyzing alpha and beta diversity of post-transplantation microbiomes revealed no significant difference between the patients who received ciprofloxacin prophylaxis and those who did not (Figure 4 and Supplementary Figure S2B and D). Dynamics of the microbial community relative abundance at the family and genus level are shown in Supplementary Figure S3. Taxa abundance at various post-transplantation time points was stratified by ciprofloxacin prophylaxis (Supplementary Figure S4). MaAsLin2 analysis of the post-transplantation samples did not reveal any significant difference in taxa abundance between patients who received ciprofloxacin and those who did not. LefSe analysis of the post-transplantation samples revealed differential increases in *Enterococcus saccharolyticus* and *Enterococcus gallinarum* after ciprofloxacin prophylaxis (Supplementary Figure S5). Analysis of the functional capacity of the metagenomes and the ARGs showed no significant difference in alpha or beta diversity metrics in the post-transplantation samples when stratified by ciprofloxacin prophylaxis. ALDEx2 analysis of the functional data and the ARG dataset also did not reveal any significant differential

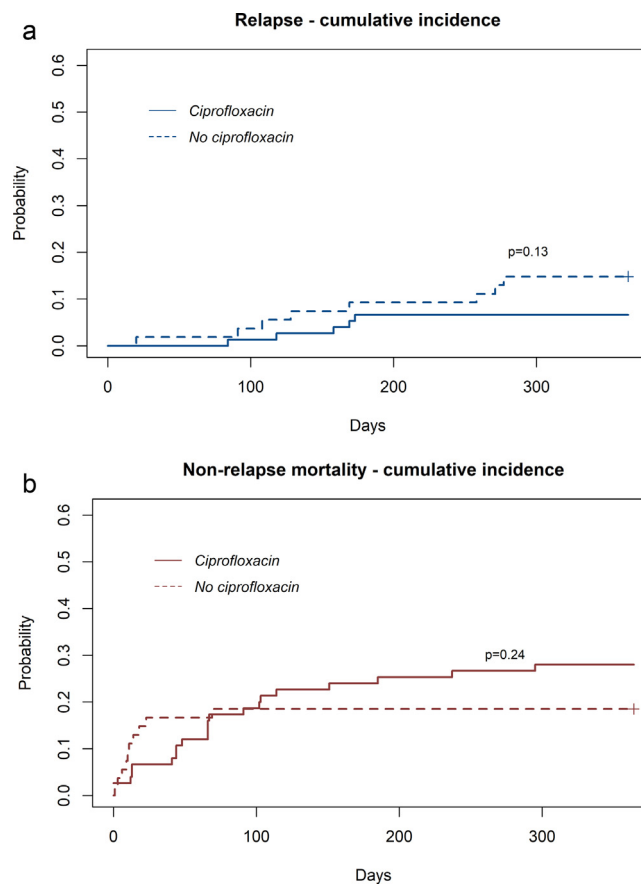


Figure 3. Competing-risk analysis for relapse versus NRM. (A) Cumulative incidence of relapse grouped by ciprofloxacin prophylaxis (solid line, ciprofloxacin; dashed line, no ciprofloxacin). (B) Cumulative incidence of NRM grouped by ciprofloxacin prophylaxis (solid line, ciprofloxacin; dashed line, no ciprofloxacin) within the first year after allo-HSCT. Results from Gray's test for each outcome are depicted.

Post-transplant alpha diversity metrics grouped by ciprofloxacin exposure

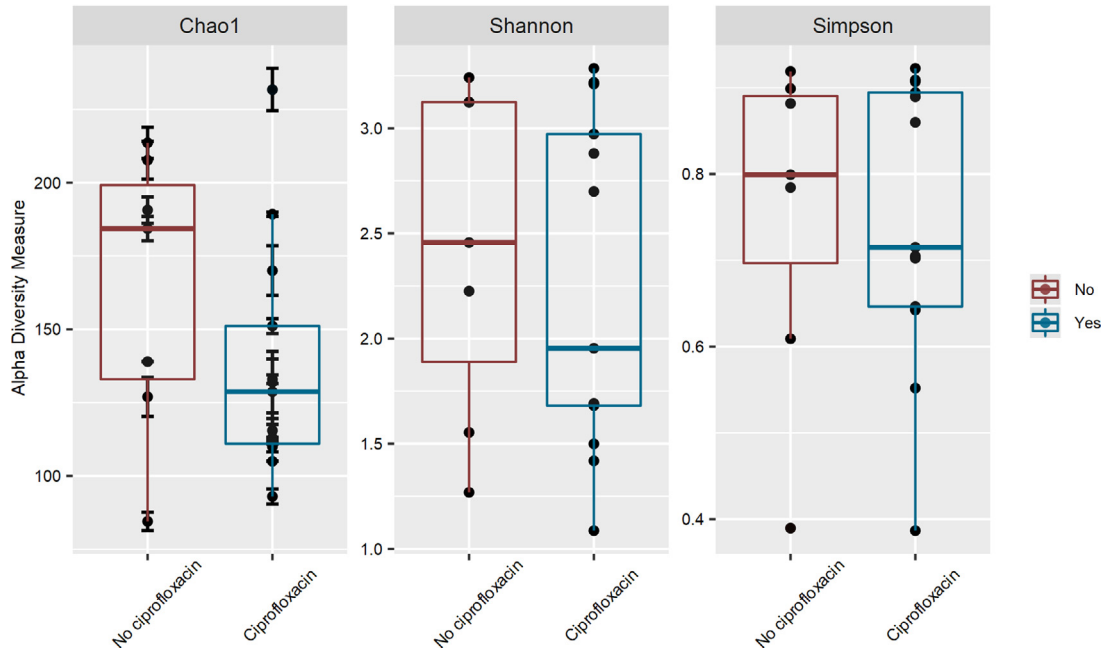


Figure 4. Post-allo-HSCT alpha diversity metrics. Chao1 (left), Shannon (middle), and Simpson (right) indices of post-allo-HSCT samples compared between patients who received ciprofloxacin prophylaxis and those who did not. *P* values were not significant for all comparisons. The central line represents the median. Boxes represent the interquartile range (IQR), and whiskers are the furthest point within 1.5 times the IQR above or below the IQR. The additional whiskers in Chao1 represent the 95% confidence interval of the metric.

abundance of genes between patients who received ciprofloxacin prophylaxis and those who did not.

DISCUSSION

In this study, omitting prophylactic ciprofloxacin during allo-HSCT was not associated with a significant difference in lower GI GVHD rates. This result was supported by microbiome analysis from stool samples collected from a subset of patients periengraftment that revealed no significant differences in alpha and beta diversity between the study groups. Furthermore, other clinical outcomes, such as the incidence of overall acute GVHD, OS, NRM, relapse rate, and BSI events, did not differ between patients who received prophylactic ciprofloxacin and those who did not.

Consistent with our findings, Rashidi et al. [23] did not see a wide effect of levofloxacin prophylaxis on alpha diversity during allo-HSCT, finding specific influences only on the abundance of *Parabacteroides* and *Blautia* [23]. Furthermore, Ziegler et al. [24] evaluated the effects of levofloxacin and broad-spectrum β -lactam antibiotics on the gut microbiome of patients with hematologic disorders and found inferior levofloxacin disruption of the gut microbiome compared with that inflicted by the β -lactam antibiotics. This finding may support our observation of the limited effect of ciprofloxacin prophylaxis on the gut microbiome in the context of extensive use of broad-spectrum β -lactam antibiotics.

Numerous studies have reported an association between intestinal microbiota diversity and clinical outcomes. According to Taur et al. [12], who analyzed stool samples from 80 patients, mortality outcomes were significantly worse in those with decreased intestinal microbial diversity. This finding is consistent with results from a large prospective observational study that tested samples from more than 1300 patients and reported a correlation between increased microbiome diversity periengraftment and better survival with a lower NRM [9].

Furthermore, low bacterial alpha diversity at day 10 post-transplantation was shown to correlate with increased risk for acute GVHD within 1 month post-transplantation [11]. However, our study did not find any differences in clinical outcomes or microbiome diversity between the study groups. Although this may represent correlating findings, the small subcohort analyzed for microbiome sequencing preclude us from stating so unequivocally.

Beside diversity, the abundance of specific taxa also was found to be associated with transplantation outcomes. *Enterococcus* genus dominance has been associated with increased NRM, GVHD, and infections [25–27]. The relative abundance of *Enterococcus* can increase during chemotherapy and also is affected by antibiotic treatment. Imipenem/cilastatin antibiotic administration was associated with higher *Enterococcus* abundance in allo-HSCT recipients and with higher rates of GVHD-related mortality after 5 years [10,28]. Increased *Enterococcus* abundance was correlated with poor overall survival, with 83.9% 2-year survival in patients with a relative abundance of *Enterococcus* <1%, compared with 47.6% in those with an abundance $\geq$ 1% [27]. In our study, several *Enterococcus* species were differentially enriched in patients who received ciprofloxacin prophylaxis on LeFSe analysis. Interestingly, the abundance of specific *Enterococcus* species, including *E. gallinarum*, in allo-HSCT recipients has been previously associated with favorable outcomes post-HSCT [29,30]. In one study, pretransplantation colonization with intrinsically vancomycin-resistant enterococci was associated with favorable OS and lower NRM compared with controls [29]. The high abundance of *E. gallinarum* may explain in part why no dismal effect was associated with ciprofloxacin prophylaxis in our cohort. However, this finding did not reflect significantly on the lower GI GVHD rate or other clinical outcomes, perhaps because of the small sample size.

Despite the advantages of gut decontamination and its wide implementation, it may have an unfavorable effect on

microbiome composition and outcomes [31–33]. In a study of 500 patients, the absence of antibiotic prophylaxis was correlated with better survival (adjusted hazard ratio, 1.61; $P < .001$) and a reduced incidence of grade 2–4 GI GVHD compared with antibiotic prophylaxis administration (20.7% versus 10.8%; $P = .003$) [14]. Alternative antibiotic prophylaxis regimens for this purpose may have different impacts on microbiome diversity. Rifaximin prophylaxis, even when followed by broad-spectrum antibiotic treatment, was associated with a higher Shannon index and better maintenance of microbiome diversity compared with ciprofloxacin/metronidazole prophylaxis regimens, as reflected in a lower rate of grade 2–4 GI GVHD, lower NRM, and higher OS in the rifaximin group [13]. Furthermore, early administration of broad-spectrum antibiotics for neutropenic fever occurring pretransplantation was associated with decreased microbiome diversity and diminished abundance of commensal Clostridiales compared with later antibiotic initiation or no antibiotic treatment. This translated to a significantly increased incidence of transplantation-related mortality and GVHD-related mortality in the early treatment group and was not affected by the fact that all patients received antibiotic prophylaxis, 73% of them with a ciprofloxacin-based regimen [34]. This in turn may explain the absence of difference in alpha or beta diversity in our study, in which 98% of patients (127 of 129) were exposed to broad-spectrum antibiotics during allo-HSCT, with most of them (85.8%; 109 of 127) receiving this treatment by day 14 post-transplantation. This may have had a dramatic effect on the microbiome collected post-transplantation at a median of 16 days post-transplantation, irrespective of prior ciprofloxacin prophylaxis administration.

We did not observe any significant difference in the rate of any BSI by day 40 post-transplantation or of gram-negative bacteria and *Enterococcus* BSI by 14 days post-transplantation. Although these findings might appear to favor stopping fluoroquinolone prophylaxis in the setting of allo-HSCT, they should be interpreted with caution. A seminal meta-analysis showed a reduced rate of infections and even reduced mortality with systemic antibiotic prophylaxis [3,4]; however, the analysis was not exclusive for allo-HSCT recipients and included trials from almost a decade ago, and thus their findings might not reflect the current epidemiology of BSI in this patient population. Furthermore, the debate for and against prophylaxis has reemerged recently [5,7] because of data demonstrating an increased incidence of antibiotic-resistant bacteria isolates from BSI [35–39], whereas the overall rate of gram-negative bacteria BSI was decreased with fluoroquinolone prophylaxis [40]. These data suggest the need to reevaluate fluoroquinolone prophylaxis in terms of its effect on gram-negative bacteria BSI in larger prospective trials.

Our study has some limitations, starting with its retrospective design with a relatively small cohort size. The lack of the anticipated effect of reducing the lower GI GVHD rate in the no ciprofloxacin group may be attributed to the high use of concomitant broad-spectrum antibiotics, which decreased microbial diversity in both groups. Overall, the microbiome analysis of the samples obtained, although small and underpowered, was in line with and supports our clinical results. In addition, we did not match participants between the groups, so there potentially could be unmeasured confounders that mask the differences between the groups. Nonetheless, there were no evident between-group differences in the variables that were analyzed (Table 1). Finally, the different allo-HSCT protocols and patient mix in other institutions might affect the generalizability of our results.

In summary, omitting prophylactic ciprofloxacin in patients undergoing allo-HSCT did not affect the rate of GI GVHD and was not associated with worse clinical outcomes. However, this important matter should be evaluated further, preferably in prospective multicenter studies with larger cohorts, supported with microbiome analysis.

ACKNOWLEDGMENTS

Financial disclosure: All authors report no financial relationships to be disclosed.

Conflict of interest statement: There are no conflicts of interest to report.

Study funding: This study was partially funded by the MAL-KAM grant for supporting research among residents, given by Rambam Health Care Campus.

Authorship statement: Guarantor of article: H.B.-Y., H.D., I. H., I.G., M.P. and S.R. collected the data and stool samples; H. B.-Y. performed the microbiome and statistical analysis; H.D., H.B.-Y., and I.H. wrote the manuscript; and H.D., H.B.-Y., I.H., and T.Z. contributed to the study design and analysis and critically reviewed the manuscript. All authors approved the final version of the article, including the authorship list.

SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.jtct.2021.12.012.

REFERENCES

- Gratwohl A, Brand R, Frasson F, et al. Cause of death after allogeneic hematopoietic stem cell transplantation (HSCT) in early leukaemias: an EBMT analysis of lethal infectious complications and changes over calendar time. *Bone Marrow Transplant*. 2005;36:757–769.
- Taplitz RA, Kennedy EB, Bow EJ, et al. Antimicrobial prophylaxis for adult patients with cancer-related immunosuppression: ASCO and IDSA clinical practice guideline update. *J Clin Oncol*. 2018;36:3043–3054.
- Kimura S, Akahoshi Y, Nakano H, et al. Antibiotic prophylaxis in hematopoietic stem cell transplantation: a meta-analysis of randomized controlled trials. *J Infect*. 2014;69:13–25.
- Gafer-Gvili A, Fraser A, Paul M, et al. Antibiotic prophylaxis for bacterial infections in afebrile neutropenic patients following chemotherapy. *Cochrane Database Syst Rev*. 2012;1: CD004386.
- Slavin MA, Worth LJ, Seymour JF, Thursky KA. Better sepsis management rather than fluoroquinolone prophylaxis for patients with cancer-related immunosuppression. *J Clin Oncol*. 2019;37:1139–1140.
- Reply to Taplitz RA, Kennedy EB, Flowers CR, Slavin MA, et al. *J Clin Oncol*. 2019;37:1140–1141.
- Horton LE, Haste NM, Taplitz RA. Rethinking antimicrobial prophylaxis in the transplant patient in the world of emerging resistant organisms—where are we today? *Curr Hematol Malig Rep*. 2018;13:59–67.
- Beelen DW, Elmaagacli A, Müller KD, Hirsch H, Schaefer UW. Influence of intestinal bacterial decontamination using metronidazole and ciprofloxacin or ciprofloxacin alone on the development of acute graft-versus-host disease after marrow transplantation in patients with hematologic malignancies: final results and long-term follow-up of an open-label prospective randomized trial. *Blood*. 1999;93:3267–3275.
- Peled JU, Gomes ALC, Devlin SM, et al. Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation. *N Engl J Med*. 2020;382:822–834.
- Staffas A, Burgos Da Silva M, van den Brink MR. The intestinal microbiota in allogeneic hematopoietic cell transplant and graft-versus-host disease. *Blood*. 2017;129:927–933.
- Mancini N, Greco R, Pasciuta R, et al. Enteric microbiome markers as early predictors of clinical outcome in allogeneic hematopoietic stem cell transplant: results of a prospective study in adult patients. *Open Forum Infect Dis*. 2017;4:ofx215.
- Taur Y, Jenq RR, Perales MA, et al. The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. *Blood*. 2014;124:1174–1182.
- Weber D, Hiergeist A, Weber M, et al. Detrimental effect of broad-spectrum antibiotics on intestinal microbiome diversity in patients after allogeneic stem cell transplantation: lack of commensal sparing antibiotics. *Clin Infect Dis*. 2019;68:1303–1310.
- Routy B, Letendre C, Enot D, et al. The influence of gut-decontamination prophylactic antibiotics on acute graft-versus-host disease and survival following allogeneic hematopoietic stem cell transplantation. *Oncoimmunology*. 2016;6: e1258506.

15. Harris AC, Young R, Devine S, et al. International, multicenter standardization of acute graft-versus-host disease clinical data collection: a report from the Mount Sinai Acute GVHD International Consortium. *Biol Blood Marrow Transplant*. 2016;22:4–10.
16. McMurdie PJ, Holmes S. Shiny-phyloseq: Web application for interactive microbiome analysis with provenance tracking. *Bioinformatics*. 2015;31:282–283.
17. Oksanen J, Blanchet FG, Friendly M, et al. Vegan: Community ecology package. R package. Available at: <https://CRAN.R-project.org/package=vegan>. Accessed June 29, 2021.
18. Mallick H, Rahnavard A, McIver LJ, et al. Multivariable association discovery in population-scale meta-omics studies. *PLoS Comput Biol*. 2021;17:e1009442.
19. Fernandes AD, Macklaim JM, Linn TG, Reid G, Gloor GB. ANOVA-like differential expression (ALDEx) analysis for mixed population RNA-Seq. *PLoS One*. 2013;8:e67019.
20. Wickham H. *ggplot2: Elegant Graphics for Data Analysis*. 2nd ed. Switzerland: Springer International; 2016.
21. Scrucca L, Santucci A, Aversa F. Competing risk analysis using R: an easy guide for clinicians. *Bone Marrow Transplant*. 2007;40:381–387.
22. Therneau TM. A package for survival analysis in R. R package version 3.2-11. 2021. Available at: <https://cran.r-project.org/package=survival>. Accessed June 29, 2021.
23. Rashidi A, Kaiser T, Holtan SG, et al. Levaquin gets a pass. *Biol Blood Marrow Transplant*. 2020;26:778–781.
24. Ziegler M, Han JH, Landsburg D, et al. Impact of levofloxacin for the prophylaxis of bloodstream infection on the gut microbiome in patients with hematologic malignancy. *Open Forum Infect Dis*. 2019;6:ofz252.
25. Taur Y, Xavier JB, Lipuma L, et al. Intestinal domination and the risk of bacteremia in patients undergoing allogeneic hematopoietic stem cell transplantation. *Clin Infect Dis*. 2012;55:905–914.
26. Holler E, Butzhammer P, Schmid K, et al. Metagenomic analysis of the stool microbiome in patients receiving allogeneic stem cell transplantation: loss of diversity is associated with use of systemic antibiotics and more pronounced in gastrointestinal graft-versus-host disease. *Biol Blood Marrow Transplant*. 2014;20:640–645.
27. Kusakabe S, Fukushima K, Yokota T, et al. Enterococcus: a predictor of ravaged microbiota and poor prognosis after allogeneic hematopoietic stem cell transplantation. *Biol Blood Marrow Transplant*. 2020;26:1028–1033.
28. Shono Y, Docampo MD, Peled JU, et al. Increased GVHD-related mortality with broad-spectrum antibiotic use after allogeneic hematopoietic stem cell transplantation in human patients and mice. *Sci Transl Med*. 2016;8:339ra71.
29. Rashidi A, Ebadi M, Shields-Cutler RR, et al. Pretransplant gut colonization with intrinsically vancomycin-resistant Enterococci (*E. gallinarum* and *E. casseliflavus*) and outcomes of allogeneic hematopoietic cell transplantation. *Biol Blood Marrow Transplant*. 2018;24:1260–1263.
30. Rashidi A, Kaiser T, Graiziger C, et al. Gut dysbiosis during antileukemia chemotherapy versus allogeneic hematopoietic cell transplantation. *Cancer*. 2020;126:1434–1447.
31. Freifeld AG, Bow EJ, Sepkowitz KA, et al. Clinical practice guideline for the use of antimicrobial agents in neutropenic patients with cancer: 2010 update by the Infectious Diseases Society of America. *Clin Infect Dis*. 2011;52:e56–e93.
32. Tomblyn M, Chiller T, Einsele H, et al. Guidelines for preventing infectious complications among hematopoietic cell transplantation recipients: a global perspective. *Biol Blood Marrow Transplant*. 2009;15:1143–1238.
33. Ullmann AJ, Schmidt-Hieber M, Bertz H, et al. Infectious diseases in allogeneic haematopoietic stem cell transplantation: prevention and prophylaxis strategy guidelines 2016. *Ann Hematol*. 2016;95:1435–1455.
34. Weber D, Jenq RR, Peled JU, et al. Microbiota disruption induced by early use of broad-spectrum antibiotics is an independent risk factor of outcome after allogeneic stem cell transplantation. *Biol Blood Marrow Transplant*. 2017;23:845–852.
35. Ogura S, Kimura M, Takagi S, et al. Characteristics of gram-negative bacteremia during febrile neutropenia among allogeneic hematopoietic stem cell transplant recipients on levofloxacin prophylaxis. *Eur J Clin Microbiol Infect Dis*. 2021;40:941–948.
36. Gjørde LI, Moser C, Sengeløv H. Epidemiology of bloodstream infections after myeloablative and non-myeloablative allogeneic hematopoietic stem cell transplantation: a single-center cohort study. *Transpl Infect Dis*. 2017;19. <https://doi.org/10.1111/tid.12730>.
37. Koh S, Yamada K, Nishimoto M, et al. Effectiveness of antibacterial prophylaxis with non-absorbable polymyxin B compared to levofloxacin after allogeneic hematopoietic stem cell transplantation. *Transpl Infect Dis*. 2015;17:647–654.
38. Blennow O, Ljungman P, Sparrelid E, Mattsson J, Remberger M. Incidence, risk factors, and outcome of bloodstream infections during the pre-engraftment phase in 521 allogeneic hematopoietic stem cell transplantations. *Transpl Infect Dis*. 2014;16:106–114.
39. Busca A, Cavecchia I, Locatelli F, et al. Blood stream infections after allogeneic stem cell transplantation: a single-center experience with the use of levofloxacin prophylaxis. *Transpl Infect Dis*. 2012;14:40–48.
40. Maurer AK, Li S, Hunter BD, et al. Ciprofloxacin prophylaxis is associated with a lower incidence of gram-negative bacteremia in patients undergoing allogeneic hematopoietic cell transplantation. *Bone Marrow Transplant*. 2020;55:2319–2321.